# Homework 3: Variogram models and kriging

Spatial statistics 2026–2027 · Professor: Dr. Lasinio

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In this homework, I return to the Wolfcamp data. I choose a variogram model, estimate its parameters by maximum likelihood, and use it for kriging. I first check the spatial trend, because a changing mean can affect the raw variogram. I also make a map of prediction uncertainty and check predictions at stations left out of the fitting process.

Based on the feedback, I now choose the covariance and trend model by cross-validated prediction error. I keep AIC as supporting evidence. I also use a separate outer validation loop to assess the complete selection procedure without using the outer test observations to choose the model.

# 1 Data and exploratory checks

## 1.1 What I am measuring

Wolfcamp contains piezometric-head measurements from 85 locations in an aquifer in Texas. The `geoR` documentation states that coordinates are in **kilometres** and heads are in **metres**. These are continuous measurements at spatial locations. I use the supplied projected coordinates for distances.

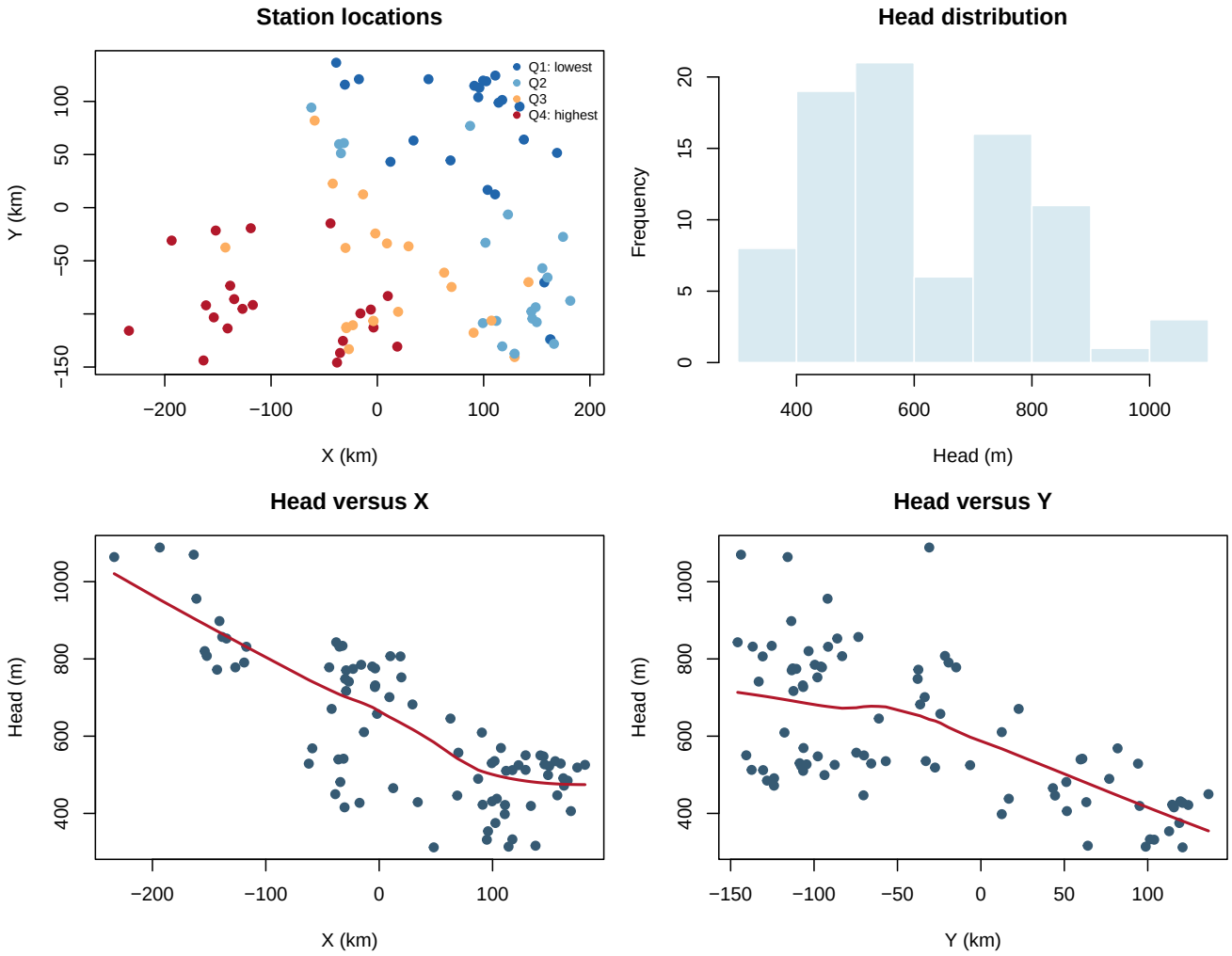
```
data("wolfcamp", package = "geoR")
dat <- data.frame(x = wolfcamp$coords[, 1],
                  y = wolfcamp$coords[, 2], head = wolfcamp$data)
checks <- data.frame(
```

```
  Stations = nrow(dat), Missing = sum(is.na(dat)),
  Repeated_locations = sum(duplicated(dat[c("x", "y")])),
  Minimum_head_m = min(dat$head), Maximum_head_m = max(dat$head)
)
stopifnot(all(is.finite(as.matrix(dat))),
          checks$Repeated_locations == 0)
knitr::kable(checks, digits = 2)
```

| Stations | Missing | Repeated_locations | Minimum_head_m | Maximum_head_m |
|---|---|---|---|---|
| 85 | 0 | 0 | 312.11 | 1088.42 |

I keep all 85 observations. There are no missing values or repeated locations. I do not treat a high or low head as an error just because it is different from nearby values.

```
quartile <- cut(dat$head, quantile(dat$head, seq(0, 1, 0.25)),
                include.lowest = TRUE, labels = FALSE)
station_col <- c("#2166AC", "#67A9CF", "#FDAE61", "#B2182B")
par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))
plot(dat$x, dat$y, asp = 1, pch = 19, col = station_col[quartile],
     xlab = "X (km)", ylab = "Y (km)", main = "Station locations")
legend("topright", c("Q1: lowest", "Q2", "Q3", "Q4: highest"),
       col = station_col, pch = 19, bty = "n", cex = 0.7)
hist(dat$head, breaks = "FD", col = "#D9EAF1", border = "white",
     main = "Head distribution", xlab = "Head (m)")
plot(dat$x, dat$head, pch = 19, col = "#365A73",
     xlab = "X (km)", ylab = "Head (m)", main = "Head versus X")
lines(lowess(dat$x, dat$head), col = "#B2182B", lwd = 2)
plot(dat$y, dat$head, pch = 19, col = "#365A73",
     xlab = "Y (km)", ylab = "Head (m)", main = "Head versus Y")
lines(lowess(dat$y, dat$head), col = "#B2182B", lwd = 2)
```




I see higher heads towards the southwest and lower heads towards the northeast. The sites are unevenly spaced. This broad pattern makes a constant mean questionable, so I examine trends before choosing the covariance model.

## 1.2 Trend and distribution

I start with ordinary linear regressions as exploratory checks. I compare a constant mean, a linear coordinate trend and a quadratic coordinate trend. Later, the likelihood models estimate the trend while allowing spatially correlated errors; I do not use these ordinary regression fits as the final kriging model.

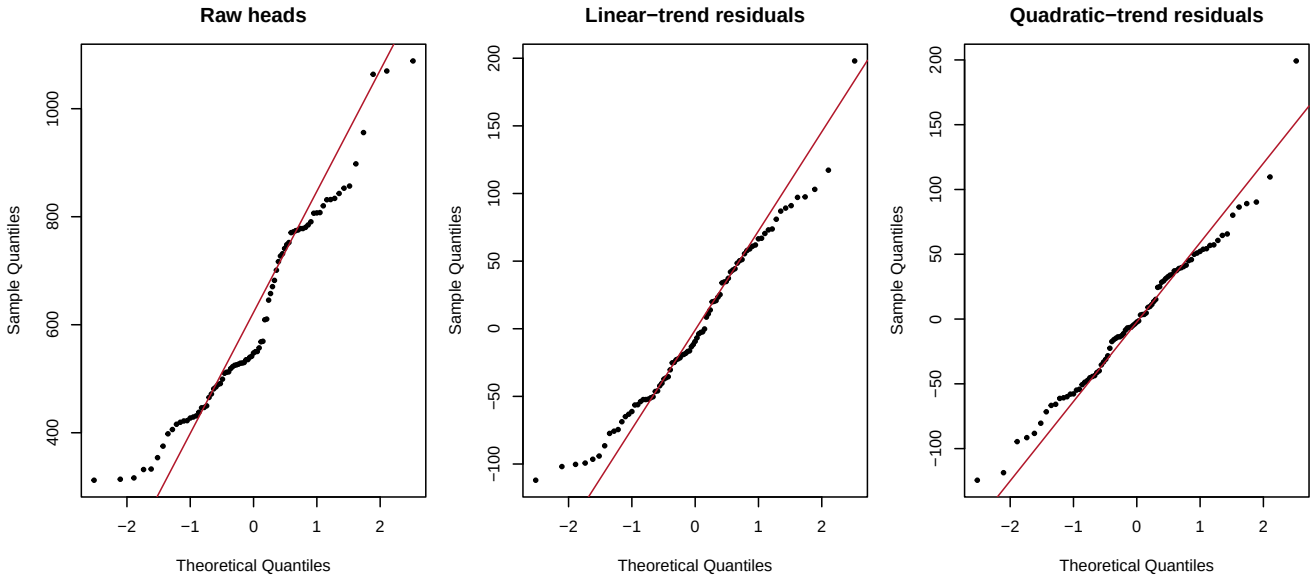
```
ols <- list(
  Constant = lm(head ~ 1, data = dat),
  Linear = lm(head ~ x + y, data = dat),
  Quadratic = lm(head ~ x + y + I(x^2) + I(y^2) + I(x*y), data = dat)
)
trend_check <- data.frame(
  Trend = names(ols),
  Adjusted_R2 = sapply(ols, function(m) summary(m)$adj.r.squared),
  Residual_SD_m = sapply(ols, function(m) sd(residuals(m)))
)
knitr::kable(trend_check, digits = 3, row.names = FALSE)
```

| Trend | Adjusted_R2 | Residual_SD_m |
|---|---|---|
| Constant | 0.000 | 186.333 |

| Trend | Adjusted_R2 | Residual_SD_m |
|---|---|---|
| Linear | 0.888 | 61.548 |
| Quadratic | 0.906 | 55.301 |

The linear trend explains much of the broad pattern: its adjusted R-squared is **0.888**. A quadratic trend improves the ordinary regression fit, but it adds three mean coefficients. I do not choose it from R-squared alone. I fit the spatial models by likelihood, then compare how well they predict observations left out of fitting.

```
par(mfrow = c(1, 3), mar = c(4, 4, 3, 1))
qqnorm(dat$head, main = "Raw heads", pch = 19, cex = 0.55)
qqline(dat$head, col = "#B2182B")
qqnorm(residuals(ols$Linear), main = "Linear-trend residuals",
       pch = 19, cex = 0.55)
qqline(residuals(ols$Linear), col = "#B2182B")
qqnorm(residuals(ols$Quadratic), main = "Quadratic-trend residuals",
       pch = 19, cex = 0.55)
qqline(residuals(ols$Quadratic), col = "#B2182B")
```




The residuals look closer to a straight Q-Q line than the raw heads, although there are some differences in the tails. I keep the original metre scale and use a Gaussian residual process as a working model. A normal-looking marginal distribution does not prove that the joint spatial distribution is Gaussian. I therefore check the predictive errors as well. No transformation is used in the fitting, validation or mapping: `lambda = 1` throughout.

## 1.3 Raw and residual variograms

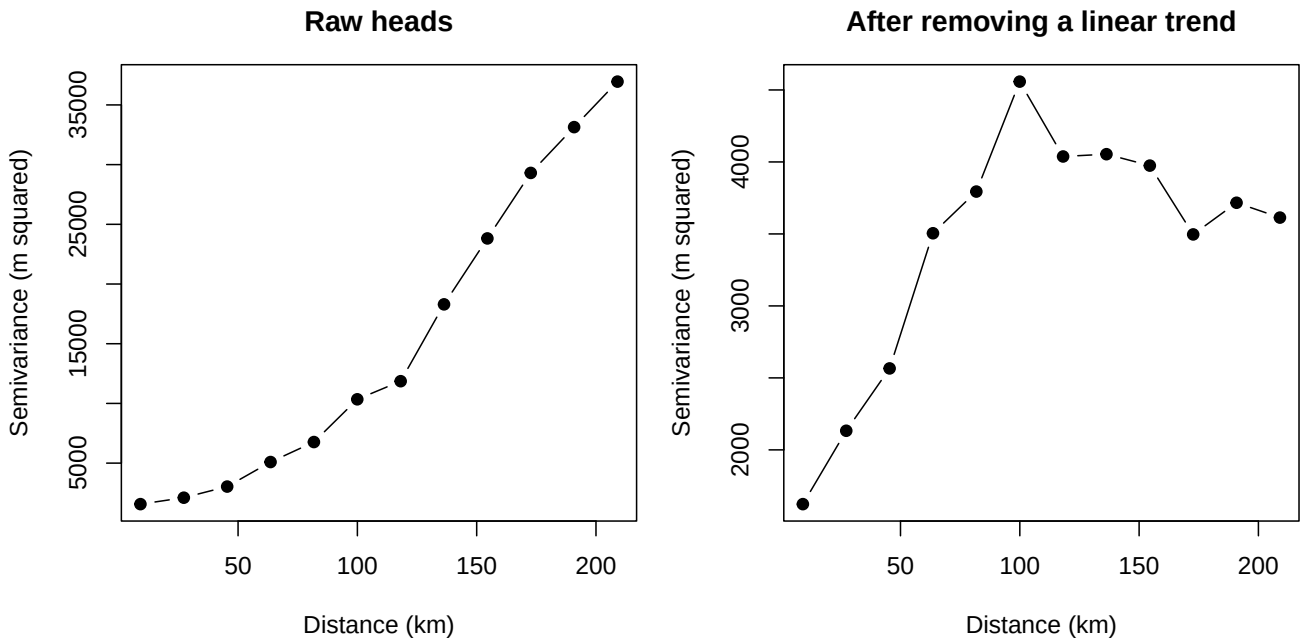
I use 12 equal-width distance groups, stopping at half the maximum station separation. I show the pair counts so that the amount of information in each group is visible. The residual variogram below removes a linear trend by ordinary least squares, as documented for `geoR::variog`.

```
cutoff <- max(dist(wolfcamp$coords)) / 2
breaks <- seq(0, cutoff, length.out = 13)
v_raw <- geoR::variog(wolfcamp, breaks = breaks, trend = "cte",
                      lambda = 1, messages = FALSE)
v_linear <- geoR::variog(wolfcamp, breaks = breaks, trend = "1st",
                         lambda = 1, messages = FALSE)
knitr::kable(data.frame(
```

```
  Lag_centre_km = v_linear$u, Pairs = v_linear$n,
  Raw_semivariance = v_raw$v, Residual_semivariance = v_linear$v
), digits = 2)
```

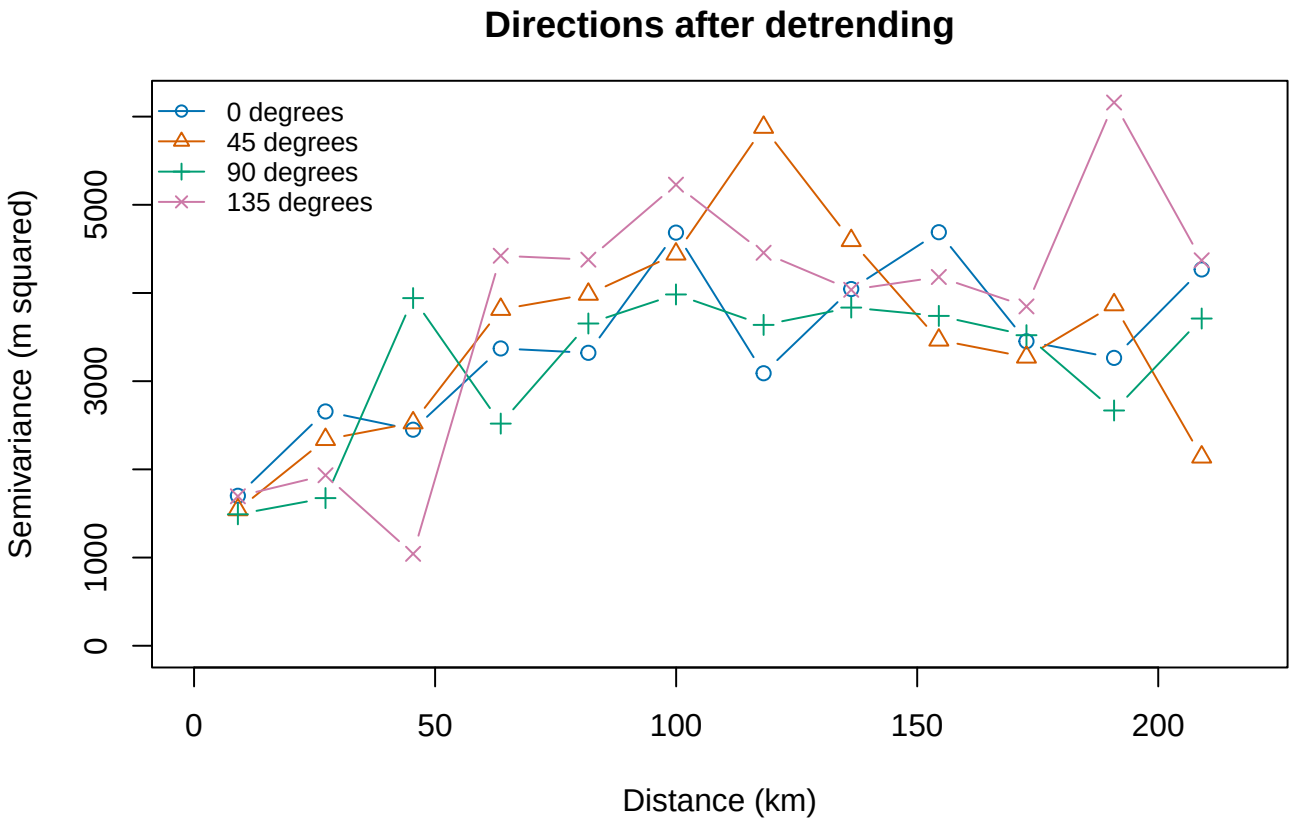
| Lag_centre_km | Pairs | Raw_semivariance | Residual_semivariance |
|---|---|---|---|
| 9.09 | 70 | 1563.74 | 1623.09 |
| 27.26 | 131 | 2104.34 | 2132.78 |
| 45.44 | 160 | 3034.78 | 2565.87 |
| 63.61 | 140 | 5090.78 | 3504.60 |
| 81.79 | 181 | 6764.60 | 3794.23 |
| 99.96 | 193 | 10343.65 | 4558.02 |
| 118.14 | 214 | 11860.59 | 4037.92 |
| 136.31 | 248 | 18296.54 | 4053.81 |
| 154.49 | 310 | 23818.00 | 3974.52 |
| 172.67 | 310 | 29305.76 | 3496.56 |
| 190.84 | 294 | 33140.75 | 3716.57 |
| 209.02 | 207 | 36953.13 | 3613.49 |

```
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
plot(v_raw$u, v_raw$v, type = "b", pch = 19,
     xlab = "Distance (km)", ylab = "Semivariance (m squared)",
     main = "Raw heads")
plot(v_linear$u, v_linear$v, type = "b", pch = 19,
     xlab = "Distance (km)", ylab = "Semivariance (m squared)",
     main = "After removing a linear trend")
```




The raw variogram keeps increasing over the displayed distances. After removing the trend, the semivariance rises over shorter distances and then roughly levels off. The panels have different vertical scales because the trend removes a large part of the variation. This supports modelling a changing mean plus a stationary residual process. It does not prove stationarity, and the residuals still show spatial structure.

```
angles <- c(0, 45, 90, 135)
directional <- lapply(angles, function(a) {
  geoR::variog(wolfcamp, breaks = breaks, trend = "1st",
    direction = a*pi/180, tolerance = pi/8, pairs.min = 10,
    lambda = 1, messages = FALSE)
})
cols <- c("#0072B2", "#D55E00", "#009E73", "#CC79A7")
par(mfrow = c(1, 1), mar = c(4, 4, 3, 1))
plot(c(0, cutoff), c(0, max(unlist(lapply(directional, function(v) v$v)))),
     type = "n", xlab = "Distance (km)",
     ylab = "Semivariance (m squared)", main = "Directions after detrending")
for (j in seq_along(angles)) {
  lines(directional[[j]]$u, directional[[j]]$v,
        type = "b", col = cols[j], pch = j)
}
legend("topleft", paste(angles, "degrees"), col = cols,
       pch = 1:4, lty = 1, bty = "n", cex = 0.8)
```




I use a tolerance of 22.5 degrees and show only directional bins with at least 10 pairs. The curves differ, especially at longer distances, but I do not see a simple, stable separation of ranges across directions. I keep an isotropic covariance model, which uses distance rather than direction. With only 85 sites, this remains a simplifying assumption; these curves do not rule out anisotropy.

# 2 Choosing a model by predictive performance

## 2.1 Model and comparison rule

My model is

$$Z(s) = m(s) + S(s) + \epsilon(s), \qquad \mathrm{Cov}\{Z(s_i), Z(s_j)\} = \sigma^2 \rho(\|s_i - s_j\|) + \tau^2 \mathbf{1}(i = j).$$

Here, `m(s)` is the trend, `S(s)` is the spatial residual process, and the independent nugget term has variance `tau^2`. The partial sill is `sigma^2`; the total sill is `sigma^2 + tau^2`. For a positive distance, the corresponding semivariogram is `tau^2 + sigma^2 * (1 - rho(h))`. At exactly zero distance, the semivariogram is zero; the nugget is its jump just to the right of zero.

I compare exponential, spherical and two Matérn models, with smoothness fixed at 1.5 and 2.5. A Matérn model with smoothness 0.5 is the exponential model under the same range convention, so the exponential candidate already covers that case. I combine these four covariance choices with the three trends.

I fit all 12 models to the original observations using **maximum likelihood (ML)** with `geoR::likfit`. I do not estimate them by fitting a curve to the empirical variogram. I use ML rather than REML for this comparison because the models have different mean structures. Their AIC values are based on the same observations and response scale.

Based on the feedback, I compare **all 12 candidates on the same five folds**. Each observation is predicted by a model fitted without its fold. I choose the lowest pooled **RMSE**, calculated from all 85 held-out errors, rather than averaging the five fold RMSEs. I also show MAE and mean error. If RMSEs are exactly tied, I use the lower MAE. I do not use AIC to shortlist candidates or override the predictive ranking.

These scores are used for model selection. The minimum is therefore not an independent estimate of the selected procedure's accuracy. Later, I repeat this comparison inside each outer training set for a separate assessment. Likelihood still estimates every fitted model's parameters; cross-validation now decides which model I use.

```
candidates <- expand.grid(
  trend = c("cte", "1st", "2nd"),
  family = c("Exponential", "Spherical", "Matern 1.5", "Matern 2.5"),
  stringsAsFactors = FALSE
)
candidates$model <- rep(c("exponential", "spherical", "matern", "matern"),
                        each = 3)
candidates$kappa <- rep(c(0.5, 0.5, 1.5, 2.5), each = 3)

fit_model <- function(g, trend, model, kappa) {
  X <- geoR::trend.spatial(trend, g)
  s2 <- mean(lm.fit(X, g$data)$residuals^2)
  # I repeat optimization from three initial range values.
  starts <- lapply(c(15, 60, 180), function(start) {
    tryCatch(geoR::likfit(g, trend = trend, cov.model = model,
      kappa = kappa, fix.kappa = TRUE,
      ini.cov.pars = c(0.8*s2, start), nugget = 0.2*s2,
      fix.nugget = FALSE, lambda = 1, fix.lambda = TRUE,
      lik.method = "ML", limits = geoR::pars.limits(phi = c(0.1, 2500)),
      messages = FALSE, control = list(maxit = 2000)),
      error = function(e) NULL)
  })
  ok <- vapply(starts, function(m) {
    !is.null(m) && m$info.minimisation.function$convergence == 0 &&
      is.finite(m$loglik)
  }, logical(1))
  if (!any(ok)) stop("No converged fit for this candidate.")
  valid <- starts[ok]
```

```
  best <- valid[[which.max(vapply(valid, function(m) m$loglik, numeric(1)))]]
  attr(best, "start_loglik") <-
    vapply(starts, function(m) if (is.null(m)) NA_real_ else m$loglik,
           numeric(1))
  attr(best, "converged_starts") <- sum(ok)
  best
}

predict_head <- function(g, locations, fit, trend, signal = TRUE) {
  geoR::krige.conv(g, locations = as.matrix(locations),
    krige = geoR::krige.control(type.krige = "OK", obj.model = fit,
      trend.d = trend, trend.l = trend, micro.scale = 0),
    output = geoR::output.control(signal = signal, messages = FALSE))
}

compare_models <- function(g, fold) {
  predictions <- variances <- matrix(NA_real_, length(g$data),
                                     nrow(candidates))
  rejected_starts <- integer(nrow(candidates))
  for (k in sort(unique(fold))) {
    test <- fold == k
    train <- geoR::as.geodata(cbind(g$coords[!test, ], g$data[!test]),
                              coords.col = 1:2, data.col = 3)
    for (j in seq_len(nrow(candidates))) {
      fit <- fit_model(train, candidates$trend[j], candidates$model[j],
                       candidates$kappa[j])
      rejected_starts[j] <- rejected_starts[j] +
        3 - attr(fit, "converged_starts")
      pred <- predict_head(train, g$coords[test, , drop = FALSE], fit,
                           candidates$trend[j], signal = FALSE)
      predictions[test, j] <- pred$predict
      variances[test, j] <- pred$krige.var
    }
  }
  stopifnot(all(is.finite(predictions)), all(is.finite(variances)),
            all(variances > 0))
  errors <- sweep(predictions, 1, g$data, "-")
  tab <- candidates[c("trend", "family")]
  tab$RMSE_m <- sqrt(colMeans(errors^2))
  tab$MAE_m <- colMeans(abs(errors))
  tab$Mean_error_m <- colMeans(errors)
  best <- order(tab$RMSE_m, tab$MAE_m)[1]
  list(row = best, table = tab, predictions = predictions,
       variances = variances, rejected_starts = rejected_starts)
}

set.seed(2026)
selection_fold <- sample(rep(1:5, length.out = nrow(dat)))
selection <- compare_models(wolfcamp, selection_fold)
# I refit on all observations after choosing the model by CV.
# The other full-data fits provide AIC and the comparison curves.
selection$fits <- lapply(seq_len(nrow(candidates)), function(j) {
  fit_model(wolfcamp, candidates$trend[j], candidates$model[j],
            candidates$kappa[j])
})
selection$fit <- selection$fits[[selection$row]]
```

```
selection$table$K <- vapply(selection$fits, function(m) m$npars, numeric(1))
selection$table$AIC <- vapply(selection$fits, function(m) m$AIC, numeric(1))
selection$table$Delta <- selection$table$AIC - min(selection$table$AIC)
chosen <- selection$fit
chosen_trend <- candidates$trend[selection$row]
chosen_family <- candidates$family[selection$row]
trend_names <- c(cte = "constant mean", "1st" = "linear trend",
                 "2nd" = "quadratic trend")
chosen_description <- paste(chosen_family, "with", trend_names[chosen_trend])
score_order <- order(selection$table$RMSE_m, selection$table$MAE_m)
runner_up <- score_order[2]
aic_best <- which.min(selection$table$AIC)
knitr::kable(selection$table[score_order,
  c("trend", "family", "RMSE_m", "MAE_m", "Mean_error_m")],
  digits = 2, row.names = FALSE)
```

| trend | family | RMSE_m | MAE_m | Mean_error_m |
|---|---|---|---|---|
| 1st | Spherical | 53.45 | 41.53 | -3.28 |
| 1st | Exponential | 53.82 | 42.16 | -5.09 |
| 1st | Matern 1.5 | 54.37 | 42.60 | -3.84 |
| cte | Spherical | 54.60 | 41.81 | -5.12 |
| cte | Exponential | 54.65 | 41.85 | -5.24 |
| 1st | Matern 2.5 | 54.68 | 42.91 | -3.61 |
| 2nd | Exponential | 55.15 | 43.28 | -4.58 |
| 2nd | Matern 1.5 | 55.58 | 43.81 | -4.03 |
| 2nd | Matern 2.5 | 55.90 | 44.17 | -4.09 |
| cte | Matern 1.5 | 55.94 | 41.61 | -1.55 |
| cte | Matern 2.5 | 56.74 | 41.56 | -1.04 |
| 2nd | Spherical | 56.76 | 43.78 | -3.99 |

In the table, `cte` means a constant mean, `1st` a linear trend, and `2nd` a quadratic trend. Every row uses the same held-out observations, and each fold has its own fitted trend and covariance parameters.

My final choice is **Spherical with linear trend**, with selection-CV RMSE **53.45 m** and MAE **41.53 m**. The next candidate is **Exponential with linear trend**, with RMSE **53.82 m**. The difference is **0.37 m**. I select the lowest RMSE, but this small difference is not proof of a general advantage. A different fold partition may change the order.

The selected family and trend remain the same as in my earlier report, but the reason for choosing them has changed. They now have the lowest prediction RMSE in a direct comparison of all candidates. Some constant-mean models also predict reasonably well on these random folds, so I do not dismiss them just because their AIC is higher.

## 2.2 AIC as supporting evidence

```
knitr::kable(selection$table[score_order,
  c("trend", "family", "K", "AIC", "Delta")], digits = 2, row.names = FALSE)
```

| trend | family | K | AIC | Delta |
|---|---|---|---|---|
| 1st | Spherical | 6 | 928.73 | 0.87 |
| 1st | Exponential | 6 | 929.73 | 1.87 |
| 1st | Matern 1.5 | 6 | 930.76 | 2.89 |
| cte | Spherical | 4 | 944.09 | 16.23 |
| cte | Exponential | 4 | 945.89 | 18.03 |
| 1st | Matern 2.5 | 6 | 930.99 | 3.12 |

| trend | family | K | AIC | Delta |
|---|---|---|---|---|
| 2nd | Exponential | 9 | 927.87 | 0.00 |
| 2nd | Matern 1.5 | 9 | 928.76 | 0.90 |
| 2nd | Matern 2.5 | 9 | 929.06 | 1.19 |
| cte | Matern 1.5 | 4 | 943.73 | 15.86 |
| cte | Matern 2.5 | 4 | 945.07 | 17.21 |
| 2nd | Spherical | 9 | 928.16 | 0.29 |

I keep the same predictive ranking in this table. `K` counts the estimated mean and covariance parameters; Matérn smoothness is fixed. The minimum-AIC model is **Exponential with quadratic trend**, with AIC **927.87**. Its prediction RMSE is **55.15 m**, compared with **53.45 m** for my selected model. My selected model has AIC **928.73**, or **0.87** units above the minimum. This gives supporting likelihood evidence for the choice, but the held-out prediction errors are the reason I select it.

This replaces my earlier rule of choosing a simpler model within two AIC units. The model with the best penalized likelihood does not have to give the smallest prediction error in this sample.

## 2.3 Selected parameters and fitted curve

```
parameter_table <- data.frame(
  Parameter = c("Nugget (m squared)", "Partial sill (m squared)",
                "Total sill (m squared)", "Spherical range (km)",
                "Nugget / total sill"),
  Estimate = c(chosen$nugget, chosen$cov.pars[1],
               chosen$nugget + chosen$cov.pars[1], chosen$cov.pars[2],
               chosen$nugget / (chosen$nugget + chosen$cov.pars[1]))
)
knitr::kable(parameter_table, digits = 3)
```

| Parameter | Estimate |
|---|---|
| Nugget (m squared) | 1236.242 |
| Partial sill (m squared) | 3328.955 |
| Total sill (m squared) | 4565.197 |
| Spherical range (km) | 122.948 |
| Nugget / total sill | 0.271 |

```
knitr::kable(data.frame(Coefficient = names(chosen$beta),
                        Estimate = as.numeric(chosen$beta)), digits = 3)
```

| Coefficient | Estimate |
|---|---|
| intercept | 620.355 |
| x | -1.326 |
| y | -1.206 |

```
knitr::kable(data.frame(
  Initial_range_km = c(15, 60, 180),
  Final_log_likelihood = attr(chosen, "start_loglik")
), digits = 4)
```

| Initial_range_km | Final_log_likelihood |
|---|---|
| 15 | -460.5162 |
| 60 | -458.3671 |
| 180 | -458.3671 |

```
stopifnot(chosen$info.minimisation.function$convergence == 0,
          chosen$cov.pars[2] > 0.1, chosen$cov.pars[2] < 2500,
          chosen$nugget >= 0, chosen$cov.pars[1] > 0)
```

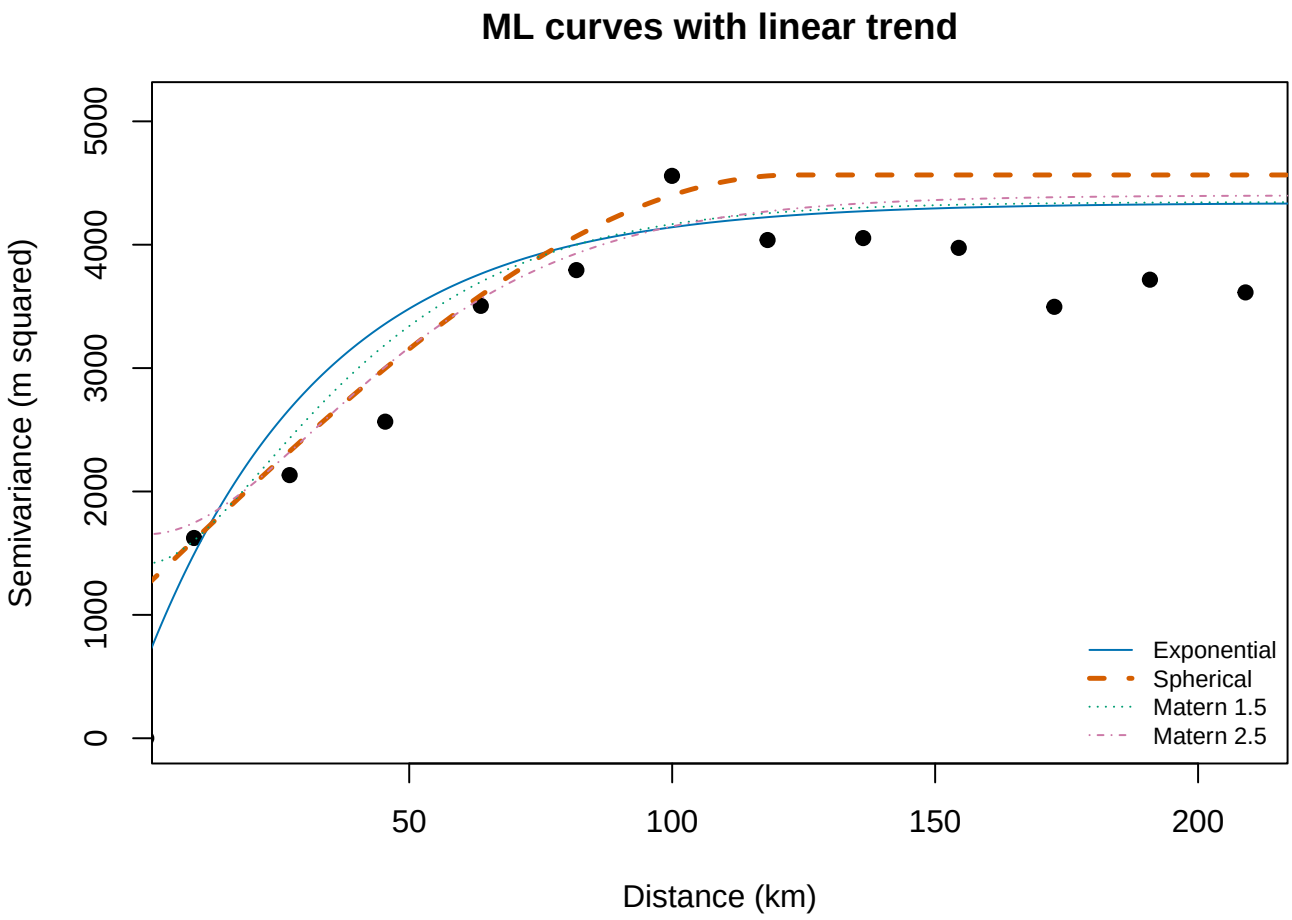
For the spherical model, correlation reaches zero at the fitted range, **122.9 km**. This is different from an exponential model, whose correlation approaches zero gradually. I therefore do not compare the `phi` values across covariance families as if they had the same meaning.

The nugget is about **27.1%** of the total sill. It can include measurement error and variation at distances smaller than those resolved by the stations. These data alone do not separate those sources. I do not choose a model just because it has the smallest nugget. Prediction error is my selection criterion; the nugget, mean structure and likelihood help me interpret the fit.

The three starting values do not all reach the same likelihood. Two reach the better solution, so using multiple starts matters here. I keep the converged fit with the greatest likelihood. The allowed range interval, 0.1 to 2,500 km, is a numerical bound; the selected estimate lies well inside it. These checks reduce the risk of accepting a poor numerical fit but do not prove that the global optimum has been found.

On some smaller training sets, an optimizer start can fail because its covariance matrix is numerically singular. I reject that start and keep the best converged fit from the remaining starts. I count rejected starts and stop if a candidate has no converged fit. I do not remove observations or silently leave a candidate out of the predictive comparison. The full-data CV comparison rejected 5 starts across all candidates and folds.

```
lag <- seq(0.001, cutoff, length.out = 300)
same_trend <- which(candidates$trend == chosen_trend)
v_selected <- geoR::variog(wolfcamp, breaks = breaks, trend = chosen_trend,
                           lambda = 1, messages = FALSE)
model_cols <- c("#0072B2", "#D55E00", "#009E73", "#CC79A7")
par(mfrow = c(1, 1), mar = c(4, 4, 3, 1))
plot(v_selected$u, v_selected$v, pch = 19,
     ylim = c(0, 1.12*max(v_selected$v, chosen$nugget + chosen$cov.pars[1])),
     xlab = "Distance (km)", ylab = "Semivariance (m squared)",
     main = paste("ML curves with", trend_names[chosen_trend]))
for (j in seq_along(same_trend)) {
  m <- selection$fits[[same_trend[j]]]
  gamma <- m$nugget + m$cov.pars[1] -
    geoR::cov.spatial(lag, cov.model = m$cov.model,
                      cov.pars = m$cov.pars, kappa = m$kappa)
  lines(lag, gamma, col = model_cols[j],
        lwd = if (same_trend[j] == selection$row) 2.5 else 1,
        lty = j)
}
points(0, 0, pch = 1)
legend("bottomright", candidates$family[same_trend],
       col = model_cols, lty = 1:4,
       lwd = ifelse(same_trend == selection$row, 2.5, 1),
       bty = "n", cex = 0.75)
```




The fitted spherical curve follows the rise over shorter distances and then levels off. It does not pass through every empirical point, especially at longer lags. The empirical plot uses an ordinary regression trend, while ML estimates the mean and covariance together from the individual observations. I use the plot as a check, not as a requirement that the fitted curve match every bin. The nearby curves also show why I should be cautious about declaring one covariance family clearly superior.

# 3 Kriging with the likelihood estimates

## 3.1 What I predict

I use the selected ML estimates in **universal kriging** because the mean contains a linear coordinate trend. In `geoR`, this is specified with `type.krige = "OK"` and the same nonconstant trend at the observation and prediction locations. Despite that argument name, the nonconstant trend makes this universal rather than constant-mean ordinary kriging.

For the main map, I predict the smooth underlying head, `m(s) + S(s)`, excluding the independent nugget term. I set `signal = TRUE` and `micro.scale = 0` explicitly. This treats all the nugget as variation excluded from that smooth target; it does not establish that the nugget is entirely instrument error.

The kriging variance allows for estimating the trend coefficients, while treating the fitted covariance parameters as fixed. It therefore does not include all the uncertainty from estimating the range, sill, nugget or choosing the model. This is likelihood-based prediction with fitted parameters, not Bayesian prediction averaged over a posterior distribution.

### 3.2 Separate assessment with nested cross-validation

I use five outer folds, with 17 stations in each fold. Inside each 68-station training set, I make a new five-fold split and compare all 12 candidates by RMSE. I then refit the selected candidate on all 68 training stations and predict the 17 untouched outer stations. The inner scores choose the model; the outer errors assess that procedure. I do not choose a model using the outer test heads or their scores.

For this check, the target is an observed head at a new location. I therefore set `signal = FALSE`, so the variance includes the nugget. The smooth-map target and the observed-validation target have different prediction variances.

```
set.seed(2026)
fold <- sample(rep(1:5, length.out = nrow(dat)))
cv_pred <- cv_var <- numeric(nrow(dat))
fold_rows <- vector("list", 5)
for (k in 1:5) {
  test <- fold == k
  train <- geoR::as.geodata(dat[!test, ], coords.col = 1:2, data.col = 3)
  set.seed(2026 + k)
  inner_fold <- sample(rep(1:5, length.out = length(train$data)))
  inside <- compare_models(train, inner_fold)
  j <- inside$row
  trend <- candidates$trend[j]
  fitted <- fit_model(train, trend, candidates$model[j], candidates$kappa[j])
  pred <- predict_head(train, dat[test, c("x", "y")],
                       fitted, trend, signal = FALSE)
  cv_pred[test] <- pred$predict
  cv_var[test] <- pred$krige.var
  fold_rows[[k]] <- data.frame(
    Fold = k, Trend = trend, Family = candidates$family[j],
    Inner_RMSE_m = inside$table$RMSE_m[j],
    Outer_RMSE_m = sqrt(mean((cv_pred[test] - dat$head[test])^2)),
    Rejected_starts = sum(inside$rejected_starts) +
      3 - attr(fitted, "converged_starts")
  )
}
stopifnot(all(is.finite(cv_pred)), all(is.finite(cv_var)), all(cv_var > 0))
errors <- cv_pred - dat$head
cv_scores <- data.frame(
  RMSE_m = sqrt(mean(errors^2)), MAE_m = mean(abs(errors)),
  Mean_error_m = mean(errors), MSSE = mean(errors^2 / cv_var),
  Coverage_95_percent = 100*mean(abs(errors) <= qnorm(0.975)*sqrt(cv_var))
)
knitr::kable(do.call(rbind, fold_rows), digits = 2, row.names = FALSE)
```

| Fold | Trend | Family | Inner_RMSE_m | Outer_RMSE_m | Rejected_starts |
|---|---|---|---|---|---|
| 1 | cte | Matern 1.5 | 52.73 | 64.74 | 3 |
| 2 | cte | Spherical | 60.93 | 38.60 | 2 |
| 3 | 1st | Exponential | 63.80 | 40.02 | 5 |
| 4 | cte | Matern 2.5 | 46.53 | 76.34 | 4 |
| 5 | 2nd | Exponential | 55.71 | 51.40 | 6 |

```
knitr::kable(cv_scores, digits = 2)
```

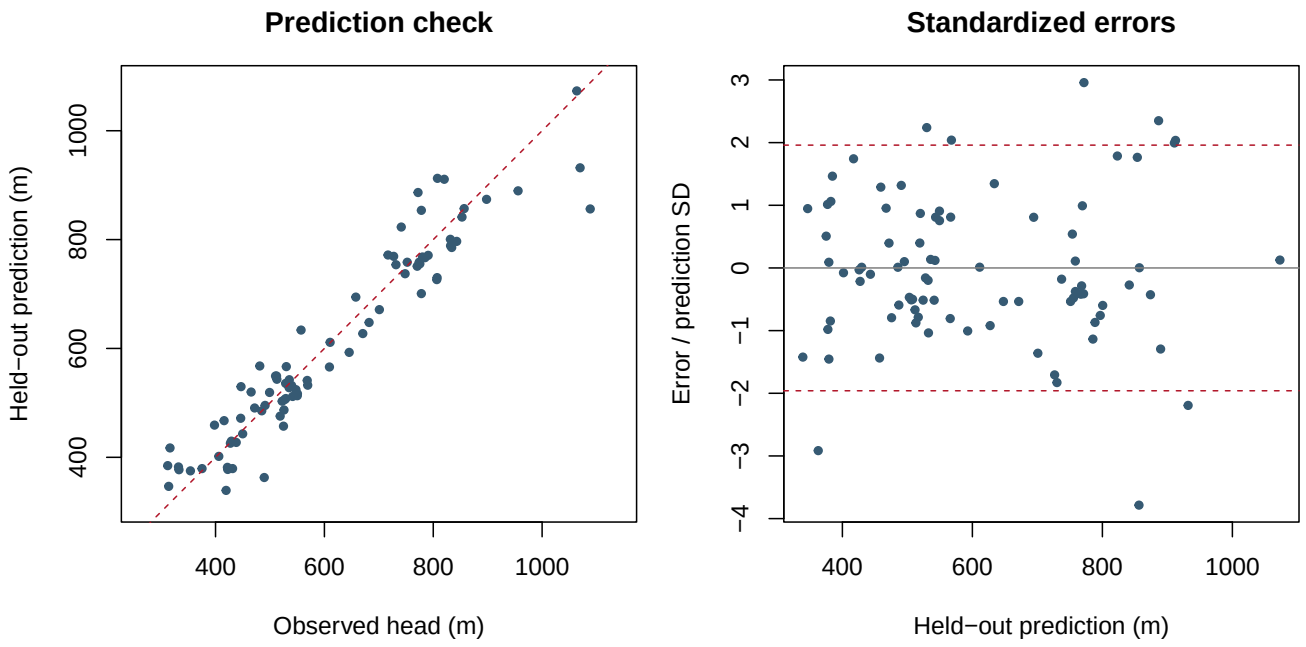
| RMSE_m | MAE_m | Mean_error_m | MSSE | Coverage_95_percent |
|---|---|---|---|---|
| 56.13 | 42.14 | -4.35 | 1.37 | 89.41 |

The pooled **outer** RMSE is **56.13 m** and MAE is **42.14 m**. These assess the predictive selection procedure. They are different from the minimum selection-CV RMSE of **53.45 m**, which helped choose the final model. The mean outer error, prediction minus observation, is **-4.35 m**. Negative values mean underprediction.

Different training sets can select different models, including a constant mean. I report these choices rather than forcing every fold to use the final full-data model. The nearby predictive scores suggest that the ranking is not completely stable.

The mean squared standardized error (MSSE) is **1.37**. A value near one would be consistent with errors on the scale of the predicted standard deviations. Values above one suggest that the prediction variances may be too small on average. The approximate 95% intervals cover **89.4%** of the outer held-out heads. I compare this with the nominal 95%, while remembering that there are only 85 predictions from one outer split. These numbers are checks of uncertainty calibration, not the criterion I use to select the model.

```
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
lim <- range(dat$head, cv_pred)
plot(dat$head, cv_pred, xlim = lim, ylim = lim, asp = 1,
     pch = 19, col = "#365A73", cex = 0.7,
     xlab = "Observed head (m)", ylab = "Held-out prediction (m)",
     main = "Prediction check")
abline(0, 1, col = "#B2182B", lty = 2)
plot(cv_pred, errors / sqrt(cv_var), pch = 19, col = "#365A73", cex = 0.7,
     xlab = "Held-out prediction (m)", ylab = "Error / prediction SD",
     main = "Standardized errors")
abline(h = 0, col = "grey50")
abline(h = c(-1.96, 1.96), col = "#B2182B", lty = 2)
```




Random folds can leave a nearby station in the training set. This check mainly describes prediction within the existing sampling network; prediction across large unsampled gaps may be harder. I have not repeated the partitions or used spatially separated folds here.

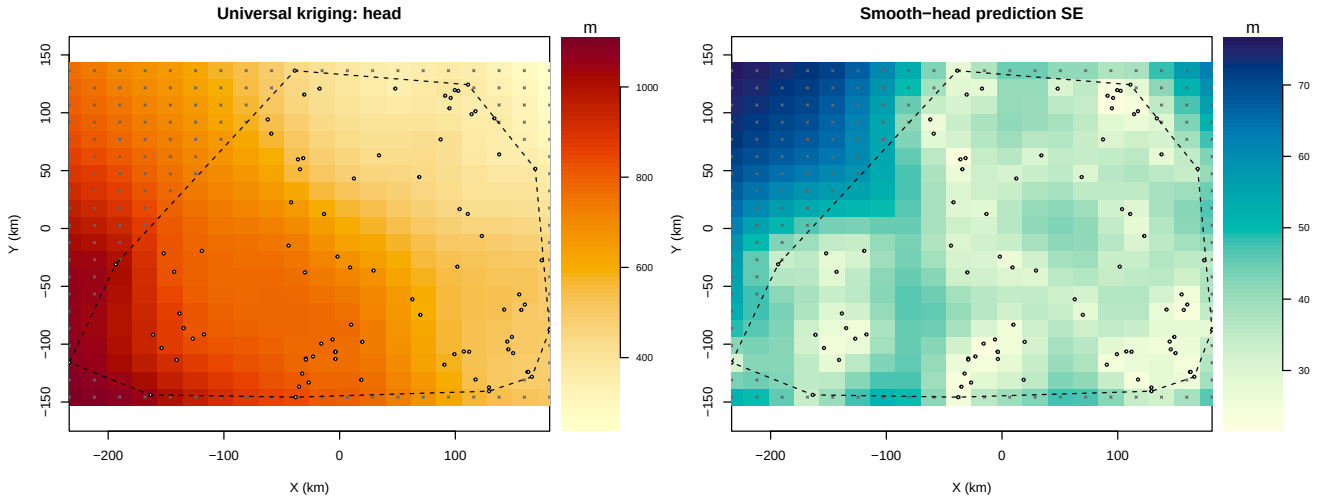
## 3.3 Final predictions on a 20 by 20 grid

I now use all 85 observations and the selected full-data model. I keep the same 20 by 20 grid definition as in Homework 1, using the observed coordinate ranges. I mark points outside the station convex hull so that the map does not hide where prediction extends beyond the sampled area.

```r
grid_x <- seq(min(dat$x), max(dat$x), length.out = 20)
grid_y <- seq(min(dat$y), max(dat$y), length.out = 20)
grid <- expand.grid(x = grid_x, y = grid_y)
prediction <- predict_head(wolfcamp, grid, chosen, chosen_trend,
                           signal = TRUE)
grid$Head <- prediction$predict
stopifnot(all(is.finite(prediction$krige.var)),
          all(prediction$krige.var >= 0), all(is.finite(grid$Head)))
grid$SE <- sqrt(prediction$krige.var)
hull <- chull(dat$x, dat$y)
grid$Inside_hull <- sp::point.in.polygon(
  grid$x, grid$y, dat$x[hull], dat$y[hull]
) > 0
grid_summary <- data.frame(
  Locations = nrow(grid), Outside_hull = sum(!grid$Inside_hull),
  Min_head_m = min(grid$Head), Max_head_m = max(grid$Head),
  Min_SE_m = min(grid$SE), Max_SE_m = max(grid$SE)
)
knitr::kable(grid_summary, digits = 2)
```

| Locations | Outside_hull | Min_head_m | Max_head_m | Min_SE_m | Max_SE_m |
|---|---|---|---|---|---|
| 400 | 140 | 241.42 | 1107.12 | 21.78 | 76.5 |

```r
draw_map <- function(values, title, colours, unit) {
  par(mar = c(4, 4, 3, 0.5))
  image(grid_x, grid_y, matrix(values, nrow = 20),
        col = colours, zlim = range(values), asp = 1, useRaster = TRUE,
        xlim = range(grid_x), ylim = range(grid_y),
        xlab = "X (km)", ylab = "Y (km)", main = title)
  points(grid$x[!grid$Inside_hull], grid$y[!grid$Inside_hull],
         pch = 4, col = "grey40", cex = 0.35)
  lines(dat$x[c(hull, hull[1])], dat$y[c(hull, hull[1])], lty = 2)
  points(dat$x, dat$y, pch = 21, bg = "white", cex = 0.5)
  par(mar = c(4, 0.2, 3, 3))
  levels <- seq(min(values), max(values), length.out = length(colours))
  image(c(0, 1), levels, matrix(rep(levels, each = 2), nrow = 2),
        col = colours, axes = FALSE, xlab = "", ylab = "")
  axis(4, las = 1, cex.axis = 0.7)
  mtext(unit, side = 3, cex = 0.8)
}
layout(matrix(1:4, nrow = 1), widths = c(1, 0.2, 1, 0.2))
draw_map(grid$Head, "Universal kriging: head",
         hcl.colors(100, "YlOrRd", rev = TRUE), "m")
draw_map(grid$SE, "Smooth-head prediction SE",
         hcl.colors(100, "YlGnBu", rev = TRUE), "m")
```




```
layout(1)
```

White circles are observed stations, the dashed line is their convex hull, and small crosses mark grid points outside it. Each panel has its own colour scale: the first shows head and the second its prediction standard error.

The predicted surface keeps the broad southwest-to-northeast decrease while allowing local changes around the observations. Predicted heads range from **241.4 to 1107.1 m**. Kriging with a trend is not restricted to the range of the observed heads, especially when predicting outside the sampled area.

There are **140 outside-hull points out of 400**. The smooth-head standard errors range from **21.8 to 76.5 m**, with larger values in poorly supported parts of the grid. Being inside the hull does not guarantee a nearby observation or an accurate model. The map also assumes that the fitted trend continues into areas where it is not directly checked.

For a new observed head at an unsampled location, the variance would include the nugget as well. With no transformation and no co-located observation, this adds `tau^2` to the smooth-head prediction variance while leaving the mean prediction unchanged. I keep this distinction in mind when interpreting the map and the held-out prediction intervals.

# 4 My AI prompt and assessment

## 4.1 Earlier exchange, before the feedback

I keep this earlier prompt and response as a record. The AIC-based selection and the 55.27 m validation result in that response belong to the previous version. They do not describe the revised selection rule or its new outer validation results.

**Earlier prompt**

> I am analysing the Wolfcamp data in R for Homework 3. I need to choose and justify a variogram model, then use kriging in the likelihood framework. Please check the data type, units, missing values and repeated locations. Examine the spatial trend and distribution before fitting the covariance. Show raw and residual variograms with distance groups and pair counts, and consider whether the patterns depend on direction. Compare a small set of reasonable models using Gaussian maximum likelihood, not just a visual fit to the variogram. Explain the parameter count, comparison rule, convergence checks, and what to do when models have similar support. Keep any needed trend or transformation consistent during fitting, prediction and validation. Make a 20 by 20 kriging map and an uncertainty map, and explain whether the target includes the nugget. If you check prediction accuracy, keep model selection inside the training data.

**Earlier recorded AI response**

> I used the 85 Wolfcamp observations and confirmed the documented coordinate units of km and head

units of m. A large-scale trend explains much of the raw spatial pattern. The detrended distribution is closer to Gaussian, so I retained the original scale and compared constant, linear and quadratic means with exponential, spherical and fixed-smoothness Matérn covariances.

The quadratic exponential model has the smallest AIC, approximately 927.87. Under the stated rule of favouring fewer parameters within two AIC units, the selected model is linear-trend spherical, with AIC approximately 928.73. Its nugget is about 1236.24 m squared, partial sill 3328.95 m squared, and spherical range 122.95 km. This is a parsimonious choice among close models, not proof that the spherical covariance is uniquely correct.

I used these ML estimates in universal kriging with the linear trend at both observed and prediction locations. The main map predicts the smooth head, while the five-fold validation predicts observed heads and includes the nugget in their variances. Repeating model selection within each training fold gives RMSE approximately 55.27 m. Covariance estimates are treated as fixed when computing kriging uncertainty, and 140 of the 400 grid points are outside the observation hull.

### 4.2 Prompt used for this revision

Based on my professor's feedback, please revise my Wolfcamp analysis so that predictive performance chooses the final trend and covariance model. Compare the main candidates on the same cross-validation folds and rank them primarily by RMSE. Keep ML for parameter estimation and use AIC only as supporting evidence. Separate the scores used to select the model from a nested validation of the selection procedure. Refit models using training observations only, retain the distinction between smooth and observed heads, and update the results and discussion using the calculations. Keep the R code and student-style wording simple.

**AI response for the revision**

I compared all 12 covariance/trend combinations on the same five folds, refitting their ML parameters within each training set. The lowest pooled RMSE selects Spherical with linear trend, with selection-CV RMSE 53.45 m. The next model has RMSE 53.82 m, so the difference should be interpreted cautiously. AIC is reported after the predictive choice and does not filter or rank the candidates for selection.

I also repeated this predictive selection inside each outer training set. The outer RMSE is 56.13 m and assesses the selection procedure; it is not the minimum score used to choose the final full-data model. The model selected for the final map is still linear-trend spherical, so its full-data likelihood estimates and kriging map remain the same apart from numerical rounding.

**My assessment**

The earlier answer was useful for separating trend and covariance, but it did not make prediction the main basis for choosing the model. Based on the feedback, I corrected that part. I can now compare all candidates' errors directly, and the code selects the minimum RMSE without using AIC. Keeping the same final model is reasonable because it also wins this predictive comparison; I do not keep it just because I used it before.

I also distinguish the selection score from the outer validation score. Choosing the smallest CV error can make that number optimistic. The outer observations are not used to choose their model, which makes the outer score more useful for assessing the whole procedure. Both checks still use random folds and depend on the available station network.

The revision prompt and response come from the same AI-assisted conversation used to update this report. They are not an independent experiment. I keep the earlier response separate so that the original claims are not silently rewritten as if they used the new procedure.

## 5 What I learned

My main choice is **Spherical with linear trend**, fitted by maximum likelihood and selected by the lowest cross-validated RMSE. Its selection score is **53.45 m**. The quadratic exponential model has a lower AIC, but a higher prediction RMSE on the same folds. I therefore keep AIC as supporting evidence and use the predictive choice for the final kriging map.

I learned that choosing a variogram is more than matching a line to empirical points. The mean model, nugget, range, likelihood and numerical fitting all matter. Based on the feedback, I now separate likelihood fitting from predictive selection, and selection error from outer assessment error. Prediction accuracy is also different from uncertainty calibration, so I report the interval coverage and MSSE alongside RMSE rather than assuming that a good predictor has perfect uncertainty estimates.

The final maps depend on approximate Gaussian residuals, an isotropic stationary covariance after detrending, and estimated parameters treated as fixed. I would be most careful with locations far from stations and outside their hull. I have not shown that this covariance family is universally best, or that this one random validation split describes prediction across large unsampled areas.

My function references are the R help pages for `wolfcamp`, `variog`, `likfit`, `cov.spatial`, `krige.conv`, `krige.control` and `output.control` in `geoR`. In particular, I use the documented distinction between ML and REML, the covariance parameter conventions, and the `signal` option for prediction.

```
packages <- c("geoR", "sp", "knitr", "rmarkdown")
knitr::kable(data.frame(
  Component = c("R", packages),
  Version = c(as.character(getRversion()),
    vapply(packages, function(p) as.character(packageVersion(p)), character(1)))
), row.names = FALSE)
```

| Component | Version |
|---|---|
| R | 4.5.0 |
| geoR | 1.9.6 |
| sp | 2.2.0 |
| knitr | 1.50 |
| rmarkdown | 2.29 |